

DAY 1 — Internet, HTML Fundamentals & Git Basics

Topics

- How the internet works
- Client vs Server
- HTTP basics
- What is a browser
- HTML structure
- Semantic HTML
- Tables
- Forms
- Input types
- Validation attributes
- Accessibility basics
- iframe, audio, video
- Git & GitHub basics

Learning Resources

HTML

- <https://developer.mozilla.org/en-US/docs/Learn/HTML>
- <https://www.freecodecamp.org/learn/2022/responsive-web-design/>
- <https://developer.mozilla.org/en-US/docs/Learn/Forms>
- <https://web.dev/learn/accessibility/>
- <https://validator.w3.org/>
- <https://www.a11yproject.com/checklist/>

Git

- <https://learngitbranching.js.org/>
- <https://www.youtube.com/watch?v=RGOj5yH7evk>

Assignment

Build a fully semantic, accessible, and polished HTML blog page from scratch and style it entirely with a dedicated CSS system.

Part 1: Structure & Accessibility (HTML)

- **Layout:** Create a webpage featuring a header/navigation, 3 article cards (each containing a title, date, excerpt, and a "read-more" link), a sidebar with a newsletter signup form, and a footer.
- **Semantics:** Use at least 6 distinct semantic HTML elements (e.g., `<header>`, `<nav>`, `<main>`, `<article>`, `<aside>`, `<footer>`).
- **Hierarchy:** Maintain a strict and logical heading hierarchy (starting at `<h1>` and descending properly to `<h2>` and `<h3>`).
- **Accessibility & Validation:** Ensure all form inputs have properly associated `<label>` elements. The final code must pass the W3C validator with 0 errors.

Part 2: Architecture & Presentation (CSS)

- **Separation of Concerns:** Keep your HTML clean—absolutely no inline styles or `<style>` tags are allowed. All styles must live in an external stylesheet.
- **Design Tokens:** Define and use CSS custom properties (variables) for at least 5 different values (e.g., primary colors, background tones, and main font families).
- **Typography & Spacing:** Implement a consistent spacing and sizing system using `rem` units exclusively throughout the stylesheet.
- **Interactivity & Polish:** Ensure the page is optimized to look professional on a 1280px wide viewport. Create smooth hover states on the article cards and clear, accessible `:focus` states on all interactive elements (links, buttons, and form inputs).

Requirements:

- Proper semantic tags
- SEO title/meta
- Contact section
- Skills section
- Project cards
- Proper indentation
- Passes W3C validator with 0 errors
- All form inputs have proper labels
- Heading hierarchy is logical (h1 → h2 → h3)
- Uses at least 6 distinct semantic elements
- Uses CSS custom properties for at least 5 values
- No inline styles on HTML elements
- Font sizes use rem units throughout
- Hover/focus states on all interactive elements

Submission:

Commit your code to the dedicated github repo before 9:00PM